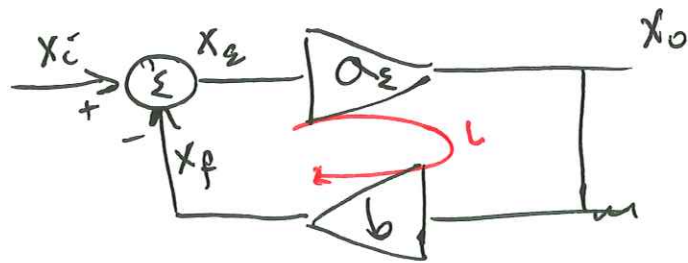


Generic Negative Feedback



$$\frac{x_o}{x_i} = A_{ideal} \left(\frac{1}{1 + 1/L} \right)$$

$$L = a_e b$$

$$A = \frac{x_o}{x_i} = \frac{a_e}{1 + a_e b}$$

$a \neq a_e$
 $\uparrow$
 not necessarily =

a_e is typically very large
 b , feedback gain. Typically ≤ 1 , ~~usually~~
 $L = a_e b$ is also typically large, > 0

$$A_{ideal} = \frac{1}{b}$$

$$\therefore \frac{1}{1 + 1/L} \geq 1$$

$$\therefore 0 \leq 1/(1 + 1/L) \leq 1$$

$$\therefore A = \frac{x_o}{x_i} = A_{ideal} \left(\frac{1}{1 + 1/L} \right) \leq A_{ideal}$$

$\rightarrow$ for $\frac{x_o}{x_i} = A_{ideal}$ $L \rightarrow \infty$
 $a_e \rightarrow \infty$
 $a \rightarrow \infty$

$$\text{Gain Error} = GE[\%] = \frac{A_{ideal} - A}{A_{ideal}} \cdot 100\% = \frac{1}{1 + L} \cdot 100\%$$

$\frac{\Delta a_e}{a_e}$ assumed $\leq 10\%$
 small

$$\frac{\partial A}{\partial a_e} = \frac{1}{(1 + a_e b)^2} = \frac{A}{a_e} \frac{1}{1 + a_e b} \rightarrow \frac{\partial A}{A} = \frac{1}{1 + a_e b} \frac{\partial a_e}{a_e}$$

$\uparrow$
 infinitesimal
 fractional
 changes

$$\frac{\Delta A}{A} \approx \frac{1}{1 + a_e b} \frac{\Delta a_e}{a_e}$$

$\Delta a_e = a_{e \text{ original}} - a_{e \text{ actual}}$
 $\Delta A = A_{\text{original}} - A_{\text{actual}}$

$$\frac{\Delta A}{A} = \frac{1}{1+L} \frac{\Delta a_e}{a_e}$$

$$L \rightarrow \infty \quad \frac{\Delta A}{A} = 0$$

$$L \rightarrow \text{very large} \rightarrow \frac{1}{1+L} \rightarrow \text{very small}$$

$$\frac{\Delta A}{A} \rightarrow \text{very small}$$

implies changes in a_e produce insignificant changes in A

$$L \rightarrow \text{small} \rightarrow \frac{1}{1+L} \rightarrow \text{approaches } 1$$

$$\frac{\Delta A}{A} \approx \frac{\Delta a_e}{a_e}$$

implies A changes drastically with a_e

really want L large!

$$L = a_e b$$

ex. 1

a_e is huge

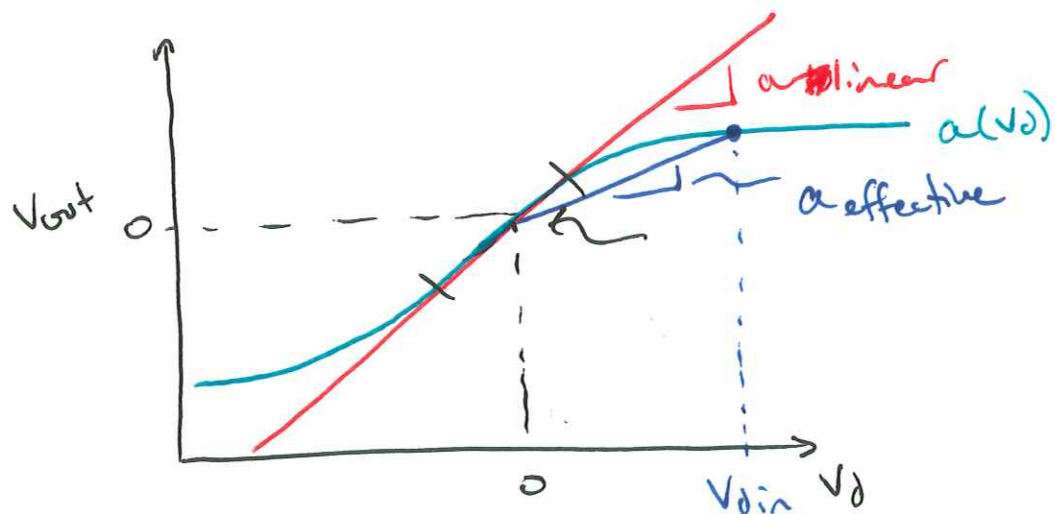
A_{ideal} is huge.

$$\text{say } A_{ideal} = 10000 = \frac{1}{b}$$

$$b = 0.00001$$

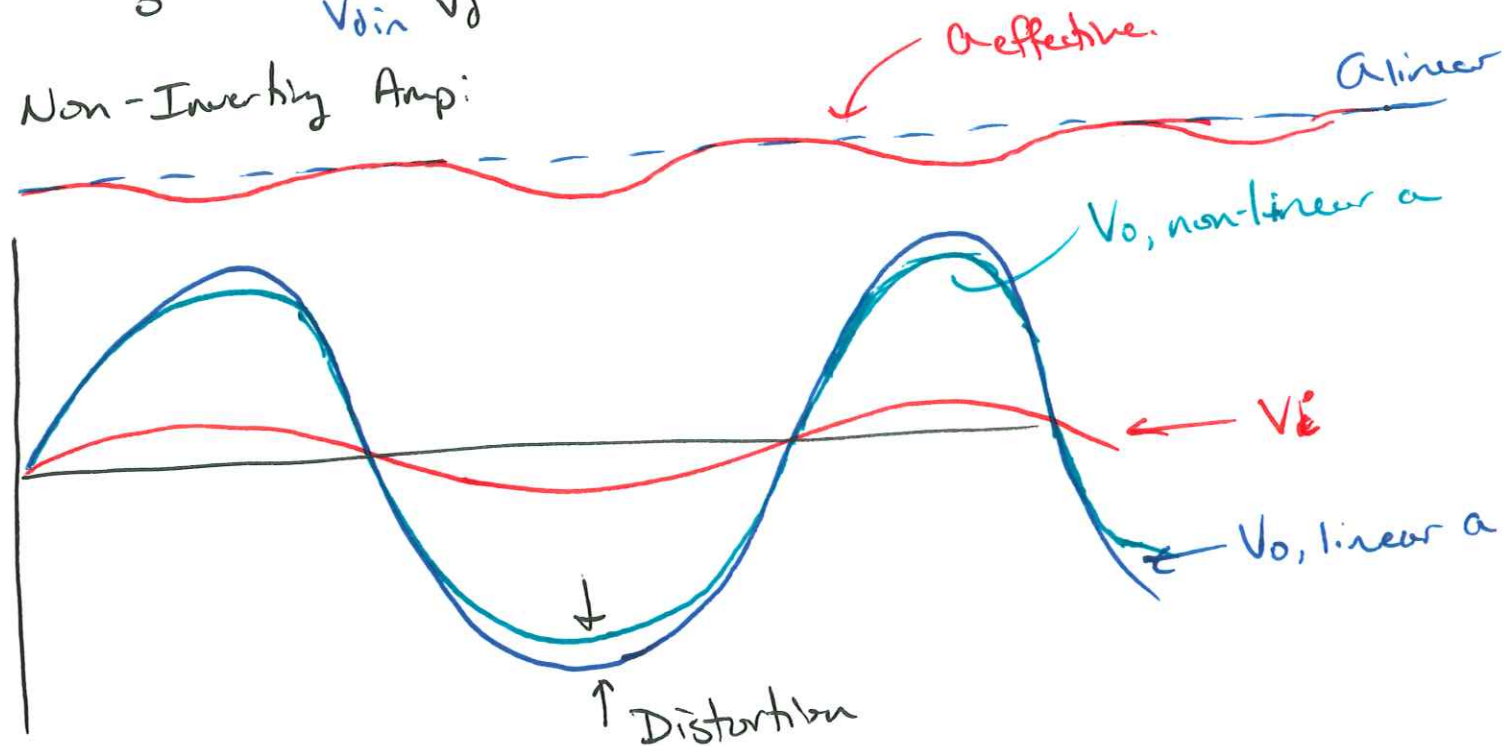
Large Ideal gains, A_{ideal} ,
significantly reduce L !

$$V_o = aV_d$$

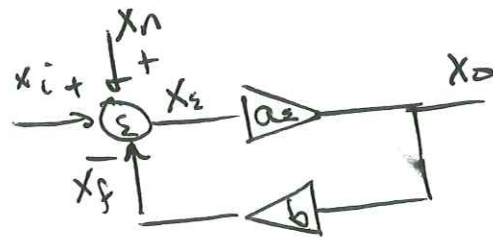


for small V_d , $a(V_d) \approx a_{linear}$

Effect for Non-Inverting Amp:



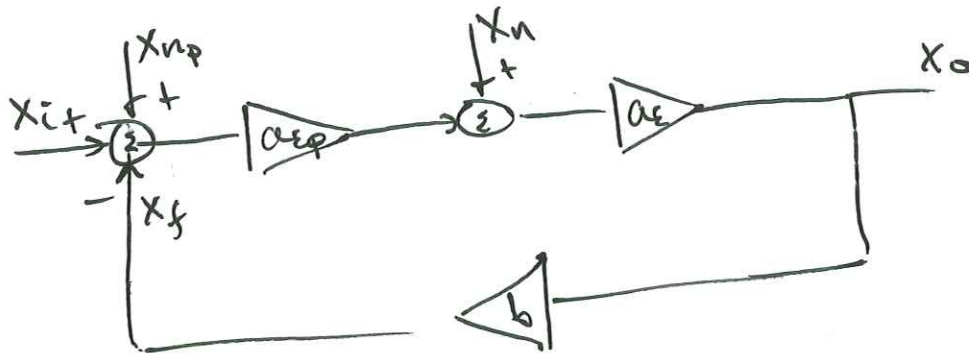
Reducing Distortion (or other fixing noisy amplifiers)



$$x_e = (x_i + x_n) - x_f$$

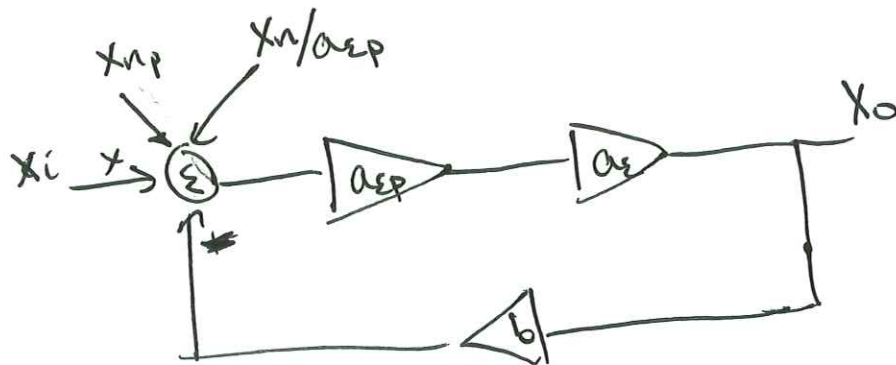
$x_n \rightarrow$ noise input to amp.

$$x_o = \frac{a_e}{1 + a_e b} (x_i + x_n)$$



$$x_o = \frac{a_{ep} a_e}{1 + a_{ep} a_e b} \left(x_i + x_{np} + \frac{x_n}{a_{ep}} \right)$$

↑
this term significantly reduces from original



a_{ep} amplifier selected to be "low noise" $\rightarrow x_{np}$ very small.

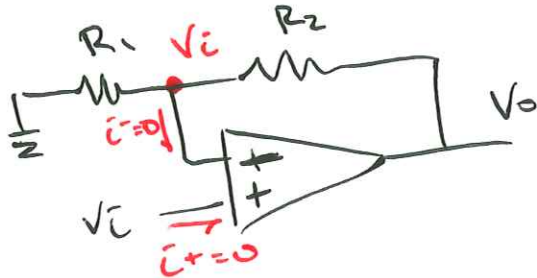
$$|x_{np}| \ll |x_n|$$

Analysis of circuits with $a \rightarrow \infty$

Also assume $i^-, i^+ = 0$

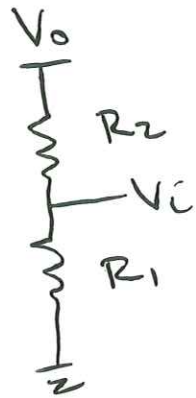
$V_o = a(V^+ - V^-) \rightarrow$ implies that $V^+ - V^- \rightarrow 0$

Assume that $V^+ = V^-$



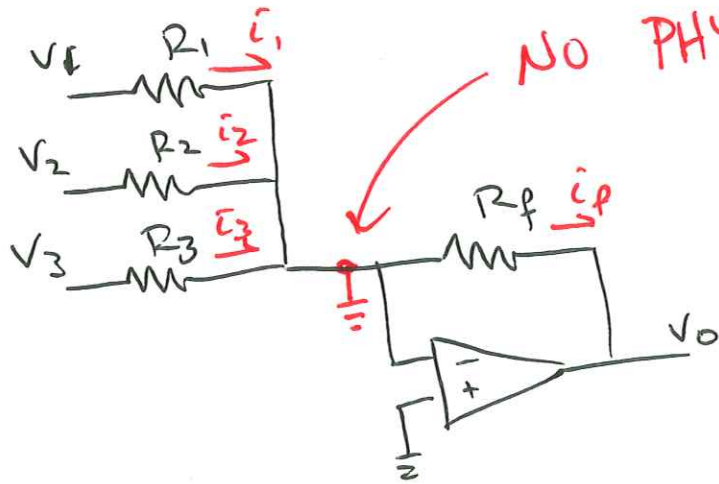
"Virtual Short Analysis"

Current: No current between nodes!



$$V_i = \frac{R_1}{R_1 + R_2} V_o \rightarrow V_o = \underbrace{\left(1 + \frac{R_2}{R_1}\right)}_{A_{ideal} = \frac{1}{b}} V_i$$

Inverting Summing Amp



NO PHYSICAL CONNECTION TO GROUND!

$$\bar{i}_1 + \bar{i}_2 + \bar{i}_3 = \bar{i}_f$$

$$\frac{V_1}{R_1} + \frac{V_2}{R_2} + \frac{V_3}{R_3} = -\frac{V_o}{R_f}$$

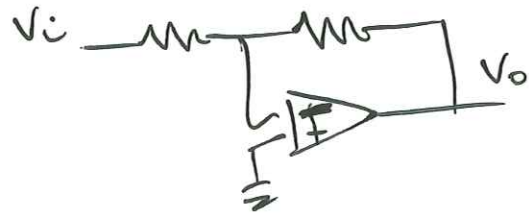
$$\bar{i}_1 = V_1/R_1$$

$$V_o = -R_f \left(\frac{V_1}{R_1} + \frac{V_2}{R_2} + \frac{V_3}{R_3} \right)$$

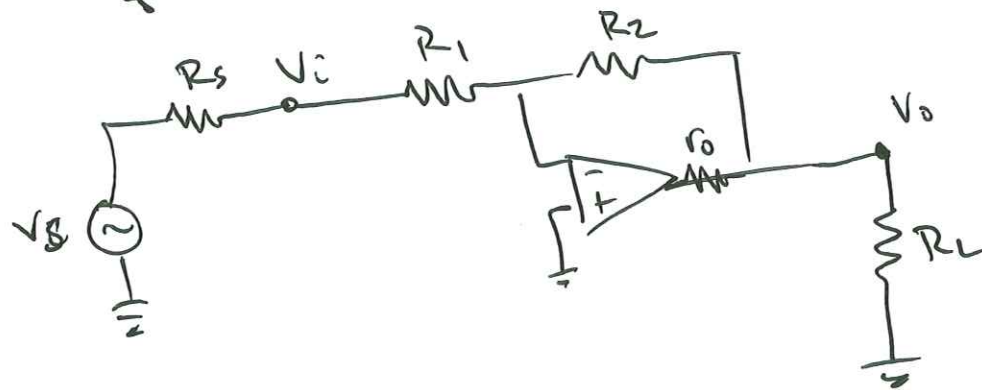
if ~~R_1, R_2, R_3~~ $R_1, R_2, R_3 = R$

$$V_o = -\frac{R_f}{R} (V_1 + V_2 + V_3)$$

Next lecture: Start with Impedances



↓ less ideal input



Ex#: Sensitivity Improvements due to loop gain increases

Have an amplifier with $a_e = 10^4$, $b = 0.1$, and some A_{orig} (can calculate)

$$\therefore L = 10^4 \cdot 0.1 = 10^3 = 1000$$

what if a_e was instead $a_e = 9 \times 10^3$

$$\Delta a_e = 10^4 - 9 \times 10^3 = 10^3 = 1000$$

$$\frac{\Delta a_e}{a_e} = \frac{10^3}{10^4} = 10\% = 0.1$$

$$\frac{\Delta A}{A} = \frac{1}{1+L} \frac{\Delta a_e}{a_e} = \frac{1}{1+1000} (0.1) = 0.01\%$$

$\approx \frac{1}{1000}$

$$\frac{\Delta A}{A} = 0.0001 \rightarrow A_{orig} - A_{new} = 0.0001 A_{orig}.$$

$$A_{new} = (1 - 0.0001) A_{orig}.$$